

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics, and Economics
Honour School of Economics and Management
Honour School of History and Economics

MICROECONOMICS

TRINITY TERM 2020

Wednesday 27 May

Opening time 09:30 (BST)

You have 4 hours to complete the paper and upload your answer file.

Please start the answer to each question on a separate page.

There are 10 questions in this paper.

*Answer **all** questions from Part A and **two** questions from Part B.*

Part A attracts 40% of the marks in total; each question in Part B attracts 30% of the marks.

In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets;

in Part B, the weight assigned to each part of the multipart question is indicated in square brackets.

The limit for essays is **1600 words**, with technical material (sensibly sized diagrams &/or equations, etc) contributing to the total at the rate of 400 words per A4 page.

Candidates may use their own calculators.

Part A

In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets

1. Consider an exchange economy with two consumers and two types of good. Consumer A has preferences represented by the utility function $u^A(x^A, y^A) = 2 \ln x^A + \ln y^A$, and an endowment of $(4, 4)$ (that is, 4 units of each good); B has preferences represented by the utility function $u^B(x^B, y^B) = \ln x^B + 2 \ln y^B$, with an endowment of $(11, 8)$ (that is, 11 units of good x and 8 units of good y).
 - (a) Explain why the endowment is not Pareto efficient.
Illustrate this situation using an Edgeworth box.
 - (b) Suppose that good y cannot be transferred between the consumers.
Find the transfer (of good x) that would result in a Pareto efficient allocation.
Mark this allocation on your diagram.
 - (c) Can the initial allocation and the allocation in part (b) be ranked using the Pareto criterion? Explain why or why not.
 - (d) Now suppose that good y can also be transferred between the consumers.
Show that the allocation that would be chosen by a social planner whose objective was to maximise the sum of the utilities of the two consumers is the same as the efficient allocation from part (b).
Is this different from the allocation that a social planner would choose if their objective was to maximise the minimum of the utilities of the two consumers?
How do you know?

[25%]

2. Two colleges, Larches and Osier, are currently equal in the academic league table, but each is considering extra coaching for their students. They simultaneously decide whether to *Maintain* current levels, *M*, or *Increase* the level of coaching, *I*, payoffs being given by:

		LARCHES	
		<i>Maintain</i>	<i>Increase</i>
OSIER	<i>Maintain</i>	0 2	−1 −4
	<i>Increase</i>	−1 −4	2 −4

- (a) Define the Nash equilibrium of a game.

Find all the Nash equilibria of the one-shot game above (in both pure strategies and mixed strategies, if any).

Calculate the expected payoffs of each college in each of the equilibria.

- (b) Explain what is meant by a *subgame perfect equilibrium* of a repeated game.

- (c) Suppose that the two colleges repeat the above stage game infinitely often.

Show that (*Maintain*, *Maintain*) in every period can be sustained as the outcome of a subgame perfect equilibrium for a discount factor sufficiently close to one.

[*Hint*: consider a grim trigger strategy with the threat of reverting to a suitable Nash equilibrium of the stage game from part (a).]

[25%]

3. (a) Define the certainty equivalent of a lottery, and explain how it can be used to determine whether or not someone should participate in a given lottery.

Janet is an expected utility maximiser who prefers more to less, and whose broad attitude to risk (risk averse, risk neutral, or risk loving) is independent of her wealth.

There are two projects. If Janet invests in project X , her final wealth will be either 6 or 30, each equally likely; if she invests in project Y , her final wealth will be either 12 or 20, again each equally likely. She is prepared to invest in one or the other, but is not allowed to invest in both.

- (b) She tells her friend Sam that she prefers Y to X , from which Sam infers that Janet is risk averse, reasoning as follows:

Consider project Z which would leave Janet with either 14 or 22, each equally likely. She would certainly prefer Z to Y , and if she were risk neutral or risk loving then she would be content with X instead of Z , and consequently prefer X to Y – but she doesn't, so she must be risk averse.

Evaluate Sam's argument.

- (c) Janet now hears that the cost of investing in project Y is going to increase, effectively reducing her final wealth. If her utility from final wealth w is $\ln w$, what is the largest cost increase Janet would find acceptable, above which she would prefer to invest in project X ?

[25%]

4. There are two types of worker: H -types with high productivity θ_H , and L -types with low productivity θ_L ; each worker knows their own type, but they have no outside option. Many risk-neutral firms compete for the services of the workers, but they cannot observe a worker's type.

- (a) Why would some of the workers like to be able to signal their type to the firms?
What condition(s) would make a signal credible?

Suppose that $\theta_H = 500$, $\theta_L = 400$, and that $3/5$ of the workers are H -types, so $2/5$ of the workers are L -types; the firms are aware of these productivity levels and proportions.

- (b) When no signal is available, what is the market equilibrium outcome?
(c) Suppose the workers can acquire an education, observable by the firms – this would cost 80 for an L -type, but only 60 for an H -type.

Consider an outcome in which H -types signal and L -types don't, and the firms pay appropriate wages.

Is this outcome a market equilibrium? Explain why or why not.

If not, find an alternative outcome, and show that it is an equilibrium.

Is the equilibrium efficient?

[25%]

Part B

5. “To decrease inequality, the government should subsidise bread and tax caviar.”

Discuss.

6. “Free trade makes everyone better off. The UK could, and therefore should, unilaterally abolish all trade tariffs.”

Discuss.

7. Some US presidential candidates advocate breaking up the so-called ‘Big Tech’ giants (FAAMG – Facebook, Amazon, Apple, Microsoft, Google), for example by unwinding Facebook’s acquisition of Instagram and WhatsApp earlier this decade.

What are the economic arguments for and against this?

8. “Insurance companies emerged because individuals in an economy have different attitudes to risk and different levels of wealth.”

Do you agree?

9. Explain the term “agency cost” in the context of Principal-Agent problems. Why does it arise?

There are a number of situations under which the agency cost is absent (or zero).

What are these situations? For each one, describe the outcome and how the Principal achieves it.

10. Consider a duopoly in which two firms produce goods that are perfect substitutes, and they compete for one period only, simultaneously choosing their production level. The cost to firm j of producing a quantity $q_j \geq 0$ is $15q_j$, for $j = 1, 2$, and the inverse demand function in the market is

$$p(q) = 75 - q \text{ where } q = q_1 + q_2.$$

- (a) Find the pure-strategy Nash equilibrium of this game.

What is the resulting price, and how much profit does each firm make? [20%]

Suppose that before the two firms play the above quantity-setting game, firm 1 can make a cost-reducing investment that has a lump-sum price of $K > 0$ and results in its cost function becoming $9q_1$. Firm 2 knows whether or not firm 1 made the investment before the firms choose how much to produce.

- (b) If firm 1 decides to invest, what will be the Nash equilibrium of the quantity-setting game?

What is the resulting price, and how much profit does each firm make (ignoring the prior investment cost of firm 1)?

What are the reasons why firm 1 earns more than in part (a) while firm 2 earns less? [30%]

- (c) Consider the two-stage game in which firm 1 chooses whether to invest or not in the first stage, and, having noted the investment decision of firm 1, the firms engage in quantity competition in the second stage.

In a subgame-perfect Nash equilibrium of the above two-stage game, what is the largest value of K that would result in firm 1 making the investment?

When K is low enough for firm 1 to make the investment, describe fully the strategies of the two firms in the subgame-perfect Nash equilibrium. [25%]

- (d) Now suppose $K = 150$, but that before the game starts, firm 2 announces that at the second stage it will ignore firm 1's investment decision and produce its quantity from the equilibrium in part (a).

If firm 1 believed the announcement, would it invest?

Should firm 1 invest – that is, is firm 2's announcement credible? Explain. [25%]